Week 8 — Activity: Thermal Refraction by Insulating Sediments

Geophysics 3A: Potential Fields & Geothermics

May 5, 2026

Purpose

To build intuition for how low-conductivity surface layers redistribute temperature gradients without changing the total heat flow.

Geological Context

Consider two continental regions with identical basal heat flow from the mantle:

- Region A: crystalline crust exposed at the surface.
- Region B: crystalline crust overlain by a thick sedimentary basin.

The sediments are dominated by shales and porous clastic rocks with lower thermal conductivity than basement rock.

Assumptions

- Steady-state heat flow.
- No internal heat production in sediments.
- Heat flow is vertical and laterally uniform.

Tasks

1. Using Fourier's law,
$$q = -k\frac{dT}{dz},$$
explain qualitatively how the temperature gradient differs between sediments and basement.
2. Sketch temperature versus depth for both regions on the same axes. Clearly label:
    - sediment thickness,
    - basement,
    - relative steepness of gradients.
3. Indicate where isotherms are compressed and where they are stretched.
4. Which region would exhibit a higher temperature at the sediment–basement interface? Explain why.

Discussion

- How does the presence of sediments affect measured geothermal gradients?
- Which region would you expect to have higher geothermal potential?
- Why might sedimentary basins be favorable geothermal targets?